Lecture 10 // Sept 24 // Instantons on S^4

Last time: We began a discussion of instantons on S^4.

To do so, we identified $\mathbb{R}^4$ with $\mathbb{H}$ and S^4 with $\mathbb{HP}^1$,

as well as the unit quaternions with S^3 and $SU(2)$.

Today, we'll wrap up our discussion of Donaldson's theorem.

First, we need to construct a connection on $\mathbb{HP}^1$. To do this, we'll recall more facts about quaternions, then slog through a computation.

Def The imaginary part of a quaternion is $\mathrm{Im}(q) = iy + jz + kw$.

Note that $\mathrm{Im}(q)$ is in $su(2)$.

Def The conjugate of a quaternion is $\bar{q} = x - iy - jz - kw$.

Viewing $\mathbb{H}$ as a subset of $\mathrm{Mat}_{2\times 2}(\mathbb{C})$, this conjugation is the conjugate transpose of a matrix, denoted as

$$q^\dagger = \begin{bmatrix} x+iy & z+iw \\ -z+iw & x-iy \end{bmatrix}^\dagger = \begin{bmatrix} \overline{x+iy} & \overline{-z+iw} \\ \overline{z+iw} & \overline{x-iy} \end{bmatrix} = \begin{bmatrix} x-iy & -z-iw \\ z-iw & x+iy \end{bmatrix}$$

Def The norm of a quaternion is $|q|^2 = q\bar{q} = \bar{q}q = x^2+y^2+z^2+w^2$

Now, recall that we constructed connections on the line bundles over $\mathbb{CP}^1$, which are

$$A = (-n)\,\frac{z\,d\bar{z}}{1+|z|^2}$$

$$F_A = (-n)\,\frac{d\bar{z}\wedge dz}{(1+|z|^2)^2}$$

We'll do a very similar construction for $\mathbb{HP}^1$. To do so we need $d\bar{q}$ and $d\bar{q}\wedge d\bar{z}$.

Def The 1-form dq is defined in each entry:

$$dq = \begin{bmatrix} dx+idy & dz+idt \\ -dz+idt & dx-idy \end{bmatrix}$$

Further, $d\bar{q} = dq^\dagger = \begin{bmatrix} dx-idy & -dz-idt \\ dz-idt & dx+idy \end{bmatrix}$

Computing $dq \wedge d\bar{q}$, we get that

$$dq \wedge d\bar{q} = \begin{bmatrix} dx+idy & dz+idt \\ -dz+idt & dx-idy \end{bmatrix} \wedge \begin{bmatrix} dx-idy & -dz-idt \\ dz-idt & dx+idy \end{bmatrix}$$

$$= \begin{bmatrix} (dx+idy)\wedge(dx-idy) + (dz+idt)\wedge(-dz+idt) & (dx+idy)\wedge(-dz-idt)\wedge(-dz+idt)\wedge(dx+idy) \\ (-dz+idt)\wedge(dx-idy) + (dx-idy)\wedge(dz-idt) & (-dz+idt)\wedge(-dz-idt) + (dx-idy)\wedge(dx+idy) \end{bmatrix}$$

$$= \begin{bmatrix} -2i(dx\wedge dy + dz\wedge dw) & -2(dx\wedge dz - dy\wedge dt) - 2i(dx\wedge dt + dy\wedge dz) \\ \overline{2(dx\wedge dz - dy\wedge dt) + 2i(dx\wedge dt + dy\wedge dz)} & \overline{-2i(dx\wedge dy + dz\wedge dw)} \end{bmatrix}$$

Note that all entries are self-dual 2 forms in the Euclidean metric.

Prop The following is a well-defined connection on $\mathbb{HP}^1$:

$$A = \frac{\mathrm{Im}(q\,d\bar{q})}{1+|q|^2} \qquad F_A = dA + A\wedge A$$

Rmk we take the imaginary part of $q d\bar{q}$ since $\mathrm{Im}(q d\bar{q}) \in \mathfrak{su}(2)$ and we want A to be a connection on an $SU(2)$ bundle.

This differs from our approach for $\mathbb{CP}^2$, since the A we defined before wasn't actually a connection on a $U(1)$ bundle.

If we wanted, we could have taken $\mathrm{im}(z d\bar{z})$ in that numerator.

But here we specifically want an $SU(2)$ connection, so A needs to be valued in the lie algebra $\mathfrak{su}(2)$.

Claude: replace $dq d\bar{q}$ with $dq \wedge dq$

Pf What follows is a long, dry, and generally unenlightening computation.

Since $\mathrm{Im}(q d\bar{q}) = \frac{1}{2}(q d\bar{q} - dq\bar{q})$ and $d|q|^2 = d(q\bar{q}) = q d\bar{q} + dq\bar{q}$,

$\mathrm{Im}(q d\bar{q}) = q d\bar{q} - \frac{1}{2} d|q|^2$.

We now wish to compute $F_A = dA + A \wedge A$.

For simplicity, we'll ignore the $\frac{1}{1+|q^2|}$ term and first compute dB and $B \wedge B$ for $B = (1+|q^2|)A$.

$$dB = d(\mathrm{Im}(q d\bar{q})) = dq \wedge d\bar{q}$$

$$B \wedge B = \mathrm{Im}(q d\bar{q}) \wedge \mathrm{Im}(q d\bar{q}) :$$

$$= (q d\bar{q} - \tfrac{1}{2} d|q|^2) \wedge (q d\bar{q} - \tfrac{1}{2} d|q|^2)$$

$$= q d\bar{q} \wedge q d\bar{q} - \tfrac{1}{2} d|q|^2 \wedge q d\bar{q} + \underbrace{q d\bar{q} \wedge \tfrac{-1}{2} d|q|^2}_{= -\left(\frac{-1}{2}\right) d|q|^2 q d\bar{q} = \frac{1}{2} d|q|^2 \wedge q d\bar{q}} + \underbrace{\tfrac{1}{4} d|q|^2 \wedge d|q|^2}_{=0}$$

$$= q d\bar{q}\, q d\bar{q}$$

we now expand, using $d\bar{q}q = d|q|^2 - \bar{q}\,dq$:

$$= q\left(d|q|^2 - \bar{q}\,dq\right) d\bar{q}$$

$$= \left(q\,d|q|^2 - q\bar{q}\,dq\right) d\bar{q}$$

$$= q\,d|q|^2 d\bar{q} \; - q\bar{q}\,dq\,d\bar{q}$$

$$= q\,d|q|^2 d\bar{q} - |q|^2 dq\,d\bar{q}$$

$$= d|q|^2 (q\,d\bar{q}) - |q|^2 dq\,d\bar{q}$$

We again expand, using that $q\,d\bar{q} = Im(q\,d\bar{q}) + \frac{1}{2}\,d|q|^2$

$$= d|q|^2 \left(Im(q\,d\bar{q}) + \tfrac{1}{2}\,d|q|^2\right) - |q|^2 dq\,d\bar{q}$$

$$= d|q|^2 Im(q\,d\bar{q}) + d|q|^2 \left(\tfrac{1}{2}\,d|q|^2\right) - |q|^2 dq\,d\bar{q}$$

To Do: figure out in-btwn steps

$$= d|q|^2 Im(q\,d\bar{q}) - |q|^2 dq\,d\bar{q}$$

Thus $B \wedge B = d|q|^2 Im(q\,d\bar{q}) - |q|^2 dq\,d\bar{q}$

All together, we have that

$$dB + B \wedge B = dq \wedge d\bar{q} + d|q|^2 Im(q\,d\bar{q}) - |q|^2 dq\,d\bar{q}.$$

To get $dA + A \wedge A$, Since $A = \frac{1}{1+|q|^2} B$, we must use the Leibniz rule: $d(A) = d(f \cdot B) = f\,dB + df \cdot B$

Thus $dA + A \wedge A = f\,dB + df \cdot B + f^2 (B \wedge B)$. Computing each part,

$$f\,dB + df \cdot B = \frac{dq \wedge d\bar{q}}{1+|q|^2} + d\left(\frac{1}{1+|q|^2}\right) Im(q\,d\bar{q})$$

$$= \frac{dq \wedge d\bar{q}}{1+|q|^2} + \frac{-d|q|^2}{(1+|q|^2)^2} Im(q\,d\bar{q})$$

$$f^2 (B \wedge B) = \frac{1}{(1+|q|^2)^2} \left(\left(d|q|^2 Im(q\,d\bar{q}) - |q|^2 dq\,d\bar{q} \right) \right)$$

Thus $dA + A \wedge A = \frac{dq \wedge d\bar{q}}{1+|q|^2} - \frac{d|q|^2}{(1+|q|^2)^2} \mathrm{Im}(q\, d\bar{q})$

$$+ \frac{1}{(1+|q|^2)^2}\left(d|q|^2 \mathrm{Im}(q\, d\bar{q}) - |q|^2 dq\, d\bar{q} \right)$$

$$= \frac{dq \wedge d\bar{q}}{1+|q|^2} - \frac{d|q|^2}{(1+|q|^2)^2} \mathrm{Im}(q\, d\bar{q})$$

$$+ \frac{1}{(1+|q|^2)^2} d|q|^2 \mathrm{Im}(q\, d\bar{q}) - \frac{1}{(1+|q|^2)^2} |q|^2 dq\, d\bar{q}$$

$$= \frac{dq \wedge d\bar{q}}{1+|q|^2} - \frac{1}{(1+|q|^2)^2} |q|^2 dq\, d\bar{q}$$

$$= \frac{(1+|q|^2)(dq \wedge d\bar{q}) - |q|^2 dq\, d\bar{q}}{(1+|q|^2)^2}$$

$$= \frac{dq \wedge d\bar{q}}{(1+|q|^2)^2} \quad \square$$

Therefore F_A is self dual wrt euclidean metric

Exercise Check A does extend to a genuine $k=1$ bundle on $\mathbb{HP}^1$. One can also figure out which bundle A should live on by calculating c_2.

You should be left with one main objection. The assertion that $dq \wedge d\bar{q}$ is self dual was wrt the metric on $\mathbb{R}^4$, but we're working on S^4! A is a SDYM cxn wrt Euc g on $\mathbb{R}^4$, so its self-dual on both charts, but we want a SDYM on S^4! What metric should we use? One such metric is the round metric on S^4.

Def TO-DO: [insert def^n of round metric]

However, on charts, it <u>doesn't</u> agree w g_{Euc}.

To resolve this, we must discuss Conformal transformations.

Conformal Transformations

Def g_0, g_1 metrics on M are conformal if $g_1 = \lambda g_0$, $\lambda > 0$.

$f: (M, g_M) \to (N, g_N)$ is conformal if $f^*(g_N)$ is conformal to g_M

Lemma If g_0, g_1 are conformal, then $*_0 = *_1$ for 2-forms on 4-mfd

Pf if $g_1 = \lambda g_0$, and $\{e_1, e_2, e_3, e_4\}$ is on. basis of T_pM wrt g_0

$$*_0(e_1 \wedge e_2) = e_3 \wedge e_4$$

$$*_1(e_1 \wedge e_2) = \lambda *_1 \left(\frac{e_1}{\sqrt{\lambda}} \wedge \frac{e_2}{\sqrt{\lambda}} \right)$$

$$= \lambda \left(\frac{e_3}{\sqrt{\lambda}} \wedge \frac{e_4}{\sqrt{\lambda}} \right)$$

$$= e_3 \wedge e_4$$

So a 2 form is SD wrt g_0 IFF -"- g_1 if $g_0 \sim g_1$.

Fact g round on S^4 is conformal to g_{euc} on each chart U_α of S^4.

eg, the steriographic projection of g_{round} is conformal to g_{euc}

Explicitly, $g_{round} = \frac{4}{(1+|q|^2)^2} g_{euc}$

Note The round metric is essentially a generalization of arc length. Consider S^1: the distance between any 2 angles θ_0, θ_1 can be measured by arc length. If we remove the point at $-\pi_S$ embed S^1 as the unit circle in $\mathbb{R}^2$, we may use stereographic proj to y-axis.

To-Do: Add Figure

Thus A is a SDYM connx wrt (S^4, g_{round}).

Another important consequence is that conformal transformations act on the SDYM moduli space.

Claim If $f : (M,g) \to (M,g)$ is conformal diffeomorphism and ∇ ∇ is a SDYM cxn wrt g, then $f^*(\nabla)$ is a SDYM cxn wrt g

∇ $\rightsquigarrow$ $f^*(\nabla)$ $\rightsquigarrow$ $f^*(\nabla)$

g $\rightsquigarrow$ change of coords $f^*(g)$ $\rightsquigarrow$ conformal! g

Thus given any ∇ a SDYM cxn on S^4, we get many more via conformal transformations. What are CTs of S^4?

we first ask (or easier q): What are CT's of S^2? Recall that these are given by mobius transformations.

Def TO-DO: [insert def^n of mobius trans on S^2]

Prop A diffeomorphism $f:(S^2, g) \to (S^2, g)$ is conformal iff it is a mobius transformation

Pf TO-DO: [insert pf]

More generally, we can view these Möbius trans. as accounting for the symmetries of S^2 and symmetries of $\mathbb{R}^2$, thought of as the image of S^2 under stereographic proj. They are:

1) rotations, Param by $SO(3)$
2) translations, —"— $\mathbb{R}^2$
3) scaling/dilation —"— $[0,\infty)$

TO-DO: Add Figure

This gives a 6-dim'l space of conformal symmetries of S^2.

Likewise, the CT's on S^4 are

1) rotations, Param by $SO(5)$
2) translations, —"— $\mathbb{R}^4$
3) scaling/dilation —"— $[0,\infty)$

Now that we've characterized the CT's of S^4, we can apply them to our SDYM connx A on (S^4, g_{round}).

Prop A is spherically symmetric

Ie up to G_2, A is invt under $SO(5)$ action.

Pf To Do: Insert Pf

This leaves us with a 5-d space of SDYM instantons by scaling & translating A. Explicitly, for $c \in \mathbb{R}^4$ and $\lambda > 0$,

To-Do: Finish this section & wrap up discussion.